
Beyond Pattern Matching: Causal Concept Representations for Strategic Decision-Making in Chess

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Abstract

We introduce CHESSCAUSALBENCH, a controlled testbed for evaluating causal representation learning in strategic decision-making. Given a fixed structural causal model over chess positions, player context, strategic choices, and semi-synthetic payoffs, the testbed defines three standardized evaluation tasks—ATE estimation accuracy, sample-efficiency curves, and robustness to nuisance-model misspecification—and supports plug-in evaluation of any covariate representation under a doubly robust estimation pipeline. Using this testbed, we empirically characterize a representation-regime structure: expert-defined concept encodings dominate at small N and moderate-to-high confounding, while learned deconfounding representations close the gap at large N . A minimal theoretical result in a linear-Gaussian SCM formalizes the variance-reduction benefit of low-dimensional sufficient encodings and anchors the empirical findings.

1 Key Project Idea

We introduce a principled testbed for evaluating causal representations in strategic decision-making, instantiated in the chess domain. The testbed is built on a fixed structural causal model (SCM) and a semi-synthetic data-generating process (DGP) with a planted ground-truth treatment effect, and it defines three standardized evaluation tasks: (T1) ATE estimation accuracy as a function of sample size N ; (T2) robustness of ATE estimates to changes in nuisance model class within a doubly robust pipeline; and (T3) sample efficiency—the minimum N required to achieve a target estimation error. The testbed accepts any covariate representation as input and is not restricted to the three we evaluate here; code and protocol will be released to allow the community to plug in novel representations.[18, 17]

Instead of operating directly on high-dimensional state spaces (e.g., raw chess boards) that entangle many factors, we propose to work in a concept space that encodes meaningful mechanisms such as piece activity, king safety, pawn structure, material balance, and space control, derived from a strong chess engine.[14, 19] In a chess domain where both raw board states and expert-defined concepts are available, we systematically compare how different representations affect the accuracy, robustness, and sample efficiency of treatment effect estimation for strategic interventions (e.g., adopting an aggressive vs. conservative strategy).[4, 18]

Concretely, we focus on two main hypotheses: (H1) that concept-level representations reduce the number of samples required to achieve a target ATE estimation error compared to raw board encodings and a learned latent representation, and (H2) that concept-level representations yield more stable ATE estimates under changes in nuisance model classes used within a doubly robust estimation pipeline. These hypotheses instantiate Tasks T1–T3 above and are evaluated under varying

confounding strength γ and noise level α , yielding a regime map of when each representation is preferable.

2 Causal Perspective

We frame the problem in terms of interventional queries rather than observational prediction.[17] The treatment T denotes a player’s strategic choice (e.g., aggressive vs. conservative) and the outcome Y is a semi-synthetic payoff constructed from game trajectories with a planted ground-truth treatment effect.[8] We explicitly construct a structural causal model (SCM) over raw board states, player context, strategic choices, and payoffs.

The target causal estimand is the conditional average treatment effect among feasible decisions,

$$\tau = \mathbb{E}[Y(1) - Y(0) \mid F = 1] = \mathbb{E}[Y \mid \text{do}(T = 1), F = 1] - \mathbb{E}[Y \mid \text{do}(T = 0), F = 1],$$

where $Y(1)$ and $Y(0)$ are potential outcomes under aggressive and conservative strategies respectively, and $F = 1$ restricts to positions where a high-quality strategic choice is available. We assume consistency (SUTVA): $Y(t)$ equals the observed outcome when $T = t$, which is justified by the fact that treatment is defined at the level of strategic classification rather than the specific move chosen.[17, 8]

Within this SCM, concept vectors are treated as a deterministic encoding of the raw state: a map $C = f(S)$ from board configurations to concept features. Under the idealized assumption that these concept features retain all information necessary for valid adjustment—i.e., that $S \perp (T, Y) \mid C, X$ holds—an adjustment set involving (C, X) would satisfy the back-door criterion whenever (S, X) does.[17, 18] We study when concept-level covariates are sufficient for identifying τ , how they affect overlap and model misspecification sensitivity in doubly robust estimation, and how their use changes the amount of data required to accurately estimate interventional quantities across different confounding regimes.[4, 12, 7]

To isolate the role of representation, we compare three covariate spaces within the same SCM: (i) a high-dimensional raw board encoding, (ii) an expert-defined concept vector, and (iii) a learned low-dimensional representation inspired by information-bottleneck and deconfounding approaches to treatment-effect estimation.[12, 7]

3 Motivations

Modern AI systems often rely on large-scale statistical pattern matching, which yields impressive predictive performance but requires vast amounts of data and struggles with counterfactual and “what-if” reasoning.[13, 2] In contrast, human decision-making appears to rely on compact libraries of concepts and causal relations, enabling rapid adaptation to new tasks and environments once core concepts are in place.[13, 10] Causal representation learning and concept bottleneck models suggest that aligning representations with underlying causal structure can improve robustness and interpretability, but there is limited empirical evidence on when interpretable, expert-defined concepts actually improve causal effect estimation and reduce data requirements.[18, 11, 1] Strategic games like chess offer a unique testbed: they provide rich, high-dimensional raw states, well-understood expert concepts, and controlled environments for defining interventions and payoffs, and they have been used before to study causal models and the evaluation of patterns in a principled way.[14, 5, 9] By rigorously comparing raw, concept-based, and learned representations in this setting, we aim to quantify how far causal, concept-level abstractions can move us away from brute-force pattern matching toward data-efficient causal reasoning in a realistic strategic domain.[18, 3, 1]

Relation to Concept Bottleneck Models and Structural Causal Variants. Concept Bottleneck Models (CBMs)[11] learn an intermediate concept layer end-to-end, enabling interpretable prediction. Recent extensions introduce causal structure into this bottleneck: Structural Causal Bottleneck Models (SCBMs) and Causally Reliable CBMs (C²BM)[1] impose explicit causal constraints between concepts and labels, improving robustness under distribution shift. These methods learn their concept representations jointly with downstream tasks and are evaluated primarily on predictive accuracy.

Our Testbed as a Complement. Unlike SCBMs and C^2 BM, which learn their bottleneck representations end-to-end, we treat the concept map as a fixed, expert-defined abstraction and use chess as a controlled testbed to study when such interpretable abstractions empirically improve *treatment-effect* estimation and sample efficiency—causal metrics rather than predictive ones.[1, 18] CHESSCAUSALBENCH thus provides the first controlled empirical testbed to evaluate whether concept bottlenecks, fixed or learned, actually improve ATE identification and how much data they require to do so. As a natural extension, one could treat the learned bottleneck of an SCBM or C^2 BM as a fourth plug-in representation $X_{SCBM} = (h_\theta(S), X)$ evaluated on Tasks T1–T3 identically to our three representations; we leave this comparison to future work.

4 Contributions

1. We introduce CHESSCAUSALBENCH, a reusable testbed for evaluating causal representations in strategic decision-making. The testbed specifies a structural causal model, a semi-synthetic DGP with controllable confounding strength γ and noise α , three standardized evaluation tasks (ATE accuracy, sample efficiency, nuisance robustness), and a plug-in interface for arbitrary covariate representations. We release code and protocol to enable community use beyond the three representations evaluated here.[17, 14]
2. We provide a minimal theoretical result in a linear-Gaussian SCM establishing that a sufficient low-dimensional encoding $C = f(S)$ reduces the variance of a DR-style ATE estimator relative to the full raw state S , under standard regularity conditions. This formalizes the variance-reduction intuition behind the empirical sample-efficiency advantage of concept representations.[4]
3. We empirically demonstrate a *representation-regime structure*: expert concept encodings are preferred at small N and moderate-to-high confounding; learned deconfounding representations are competitive at large N ; and raw board encodings underperform across the board due to positivity and propensity instability. We summarize this as a Regime Diagram over the (N, γ) grid.[4, 12, 18]
4. We illustrate how concept-level causal abstractions support structured “what-if” queries—manipulating individual concepts (e.g., king safety) while holding others (e.g., material balance) fixed—providing evidence that such representations enable modular interventional reasoning that raw states cannot support at a human-interpretable level.[14, 5, 9]
5. We validate testbed generality with a secondary structured-gridworld environment, showing that the regime structure replicates qualitatively across domains and that the SCM/DGP protocol is not chess-specific.

5 Structural Causal Model

5.1 Variables and Structural Assumptions

We work at the level of individual decision points within games. For each decision, we define the following variables:

- S : *Raw board state*. A high-dimensional encoding of the pre-move position, including piece locations, side to move, castling rights, and other rule-relevant information. This is treated as the underlying spatial state of the game.
- $C = f(S)$: *Concept-level representation*. A low-dimensional vector of expert-defined chess concepts (e.g., king safety, material balance, piece activity, pawn structure, space control), obtained via a fixed evaluation function implemented by a strong chess engine.[14, 19]
- X : *Pre-treatment player/opponent context*. A set of exogenous and meta-variables such as player and opponent Elo, color, time control, clock features, game phase, and opening code. These variables may influence both strategic choices and outcomes but are not deterministically derived from S .
- F : *Feasibility indicator*. A binary variable indicating whether the actually played move lies within a high-quality candidate set (e.g., top- k engine moves) at state S . All causal estimands are defined conditional on $F = 1$.

- *T: Treatment (strategy choice)*. A binary indicator of the player’s strategic decision at the position, e.g., $T = 1$ for an “aggressive” move and $T = 0$ for a “conservative” move, defined by a rule-based labeling of moves using non-evaluative features (captures, checks, king-zone pressure, etc.).
- *Y: Outcome*. A semi-synthetic payoff anchored in structural position features and augmented via an explicit potential-outcomes model, with a planted ground-truth average treatment effect τ^* . This allows direct measurement of estimation bias without relying on true counterfactuals.
- *U: Latent factors*. Unobserved variables (e.g., short-term psychological factors, latent style) that may influence both T and Y . These are used in sensitivity analyses but are not available for adjustment.

We consider three alternative observed adjustment sets:

$$X_{\text{raw}} = (S, X), \quad X_{\text{concept}} = (C, X), \quad X_{\text{learned}} = (Z, X),$$

where $Z = g(S, X)$ is a learned representation (e.g., information-bottleneck or deconfounding embedding) trained to preserve information relevant for treatment effect estimation.[12, 7] These three sets correspond directly to the competing back-door approximations formalized in the SCM; Z is learned under the same DGP and evaluated in the same pipeline as the other two representations.

5.2 DAG and Causal Structure

The qualitative causal structure of the decision problem is summarized by the directed acyclic graph:

$$\begin{aligned} X &\rightarrow S, & X &\rightarrow T, & X &\rightarrow Y, \\ S &\rightarrow T, & S &\rightarrow Y, & T &\rightarrow Y, \end{aligned}$$

with an additional latent parent U influencing both T and Y :

$$U \rightarrow T, \quad U \rightarrow Y,$$

and all edges restricted to the feasible-decision population $F = 1$. The concept vector $C = f(S)$ does not appear as a separate node in this graph; it is a deterministic encoding of S and introduces no new causal ancestry beyond what is already present through S . We instead treat C as an alternative covariate encoding for the adjustment set: rather than conditioning on S directly, one may condition on $f(S)$ instead. The three adjustment sets X_{raw} , X_{concept} , and X_{learned} represent three such encoding choices for the same underlying adjustment problem. The edge $X \rightarrow S$ reflects a selection mechanism rather than a direct physical cause: players with different Elo, styles, or opening preferences systematically reach different board positions, inducing a statistical dependence between X and S in the observational distribution.

Intuitively, pre-treatment context X shapes both the raw state S (through opening choices, game phase, and time control) and the player’s strategic tendencies, and it directly affects the difficulty of the position and thus the outcome. The raw spatial state S , together with X , informs the set of plausible moves and the eventual strategy choice T , while T exerts a direct causal effect on the outcome through the chosen move.

Within this SCM, the target causal estimand is:

$$\tau = \mathbb{E}[Y(1) - Y(0) \mid F = 1],$$

where $Y(1)$ and $Y(0)$ denote potential outcomes under aggressive and conservative strategies, respectively. Identification of τ from observational data requires conditional exchangeability,

$$(Y(0), Y(1)) \perp T \mid X_{\text{adj}}, F = 1,$$

for some adjustment set X_{adj} , together with positivity and consistency.[17, 8] We do not assume that any candidate adjustment set is perfectly valid in practice. Instead, we treat $X_{\text{adj}} \in \{X_{\text{raw}}, X_{\text{concept}}, X_{\text{learned}}\}$ as competing approximations to a valid back-door adjustment set and empirically compare how close each comes to satisfying the requisite ignorability condition.

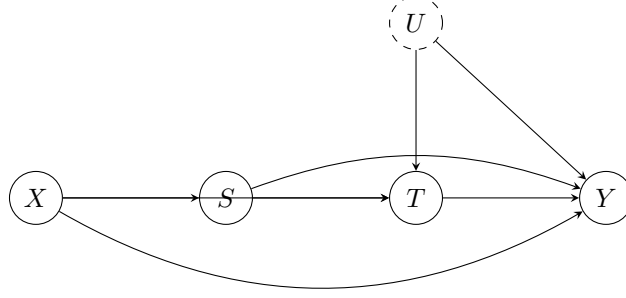


Figure 1: Structural causal model for a single chess decision. X denotes pre-treatment player/opponent context (e.g., Elo, color, time control), S the raw board state, T the strategic treatment (aggressive vs. conservative), Y the semi-synthetic payoff, and U latent unobserved factors. $C = f(S)$ is not a separate causal node: it is a deterministic encoding of S used as an alternative covariate representation in the adjustment set. We compare adjustment on (S, X) , $(f(S), X)$, and (Z, X) to assess how different encodings of state approximate a valid back-door adjustment set.

5.3 Role of Concept-Level Abstraction

Concept vectors $C = f(S)$ are a fixed, deterministic encoding of the raw state S , not a separate causal parent of T or Y . Since all information in C is already contained in S , the adjustment set (C, X) does not block any back-door paths that (S, X) does not already block—but it may approximate the true confounding structure more efficiently by discarding causally irrelevant micro-structure in S . Formally, under the idealized assumption that $S \perp (T, Y) \mid C, X$, the adjustment set (C, X) satisfies the back-door criterion whenever (S, X) does. Our experiments probe how closely this ideal holds in practice across varying confounding regimes and sample sizes.

6 Theoretical Motivation: Variance Reduction via Sufficient Encodings

To provide a formal anchor for the sample-efficiency claims, we invoke results from semiparametric efficiency theory and causal sufficient dimension reduction.[6, 15, 16]

Setup. Suppose $C = AS$, $A \in \mathbb{R}^{k \times d}$, $k \ll d$, is a sufficient statistic for S with respect to (T, Y) given X , i.e.,

$$(T, Y) \perp S \mid (C, X).$$

Under ignorability $(Y(0), Y(1)) \perp T \mid S, X$, sufficiency of C implies $(Y(0), Y(1)) \perp T \mid C, X$, so identification of τ is preserved by condition on (C, X) rather than (S, X) .

Semiparametric efficiency bound. The efficient influence function (EIF) for the ATE depends only on the σ -field generated by the covariates that are sufficient for the joint law of (T, Y) given X . [6] When C is sufficient, (C, X) generates the same σ -field for (T, Y) as (S, X) ; therefore the *semiparametric efficiency bound is preserved* when switching from (S, X) to (C, X) .

Finite-sample variance reduction. Although the asymptotic bound is identical, a DR estimator based on (C, X) estimates the nuisance functions

$$e(c, x) = P(T=1 \mid C=c, X=x), \quad \mu_t(c, x) = \mathbb{E}[Y \mid T=t, C=c, X=x]$$

rather than their high-dimensional analogues $e(s, x)$, $\mu_t(s, x)$. Because C is low-dimensional and sufficient, these nuisance functions are substantially easier to estimate well at any finite N , leading to smaller nuisance estimation error and lower finite-sample variance of the DR estimator—even though the asymptotic bound is unchanged.[15, 16]

Connection to experiments. This result predicts that at small N , concept-level adjustment should outperform raw-state adjustment in ATE accuracy, with the gap closing as $N \rightarrow \infty$ —exactly the regime structure we observe empirically. The result also implies that learned representations Z can partially recover this benefit to the extent that Z approximates a sufficient reduction of S .

Key references. Hahn (1998)[6] derives the semiparametric efficiency bound for the ATE under unconfoundedness. Ma & Zhu (2012)[15] develop semiparametric efficient estimation under sufficient dimension reduction, showing the efficiency bound is retained while nuisance variance decreases. Nabi et al. (2022)[16] extend these results to causal sufficient dimension reduction, formalizing conditions under which low-dimensional reductions preserve causal effects in semiparametric estimation.

7 Experimental Setup

All experiments target the estimand $\tau = \mathbb{E}[Y(1) - Y(0) \mid F = 1]$ as defined in the SCM above. The three adjustment sets $X_{\text{adj}} \in \{X_{\text{raw}}, X_{\text{concept}}, X_{\text{learned}}\}$ correspond directly to the competing back-door approximations formalized in the SCM.

7.1 Data and Decision Extraction

We sample decision points from classical (≥ 25 min) rated (> 1800 Elo) games obtained via the Lichess public API. For each game, we uniformly sample up to 18 positions from plies 10–120, extracting three covariate encodings per position: a raw 1152-dimensional bitboard tensor S (12 piece planes + 6 metadata planes, flattened), a 12-dimensional expert concept vector $C = f(S)$ (6 structural features computed from `python-chess` plus 6 engine-derived sub-terms), and a 32-dimensional learned embedding Z via linear deconfounding.[12] All three encodings are concatenated with a shared 6-dimensional context vector X (Elo differential, time control, move number, color, opening code, phase). The decision pool comprises $\sim 30,000$ positions.

7.2 Treatment and Outcome

The binary treatment $T \in \{0, 1\}$ is assigned via a rule-based aggression score $A(S, m)$ computed from captures, checks, king-zone pressure, sacrifices, and retreats—no engine evaluation enters T . We threshold at $A \geq 0.5$, yielding a marginal treatment rate $P(T=1) \approx 0.34$.

Outcomes are semi-synthetic with planted ground-truth ATE τ^* . The baseline potential outcome $Y(0) = \varphi(X) + \psi(C) + \gamma \cdot U + \varepsilon_0$ combines context features $\varphi(X)$ with Group A structural concepts $\psi(C)$ (material balance and king pawn shield only—no Stockfish evaluation), a latent confounder $U \sim \mathcal{N}(0, 1)$ at strength γ , and noise $\varepsilon_0 \sim \mathcal{N}(0, 0.25)$. The treated outcome is $Y(1) = Y(0) + \tau^*$ with $\tau^* = 0.3$ (homogeneous regime) or $\tau^*(X) = 0.1 + 0.3 \cdot \mathbf{1}\{\text{material} > 0\}$ (heterogeneous). Treatment-side confounding is introduced by soft-reassigning T via a logit shift $\alpha \cdot U$ with $\alpha = 0.3$.

7.3 Estimators

The primary estimator is the cross-fitted ($K=5$) Doubly Robust Learner[4] with logistic regression propensity and LightGBM outcome models. We also fit a T-Learner baseline and a DR-Learner variant with a linear (Ridge) outcome model for robustness analysis.

7.4 Experimental Grid

The full grid sweeps $N \in \{1k, 2k, 5k, 10k, 20k\}$, $\gamma \in \{0, 0.3, 0.6\}$ confounding strengths, both τ^* regimes, all three representations, all three estimator configurations, and 10 random seeds per cell (2,700 total fits). This grid is designed as a standardized protocol: any new representation can be slotted in by providing a featurization of the same positions and running the same pipeline.

7.5 Learned Representation

The learned embedding Z follows Kuang et al.[12] in a linear variant: standardize the concatenated raw features, fit a regularized logistic propensity model, project out the treatment-discriminative direction, and PCA-compress the residual to 32 dimensions. This yields a deconfounding-flavoured embedding that trains in seconds per fold with no deep-learning dependency.[7]

7.6 Secondary Environment: Structured Gridworld

To demonstrate that the testbed is not chess-specific, we instantiate the same SCM/DGP protocol on a structured 10×10 gridworld with obstacles and a goal. Variables map as follows: S is a flattened one-hot grid encoding ($d = 300$); $C = f(S)$ consists of three expert concepts (L1 distance-to-goal, obstacle density in a 3×3 neighbourhood, current quadrant index); Z is the same linear deconfounding embedding; and T is binary, assigned via a calibrated sigmoid score over both expert concepts and a hidden checkerboard parity indicator $\chi(S) = (r + c) \bmod 2$ derived from the agent cell coordinates. The treatment score includes a $+1.0 \chi_z$ term, and the outcome DGP adds a $+0.5 \chi_z$ term to $\psi(C)$ (subscript z denotes a per-feature standardized copy). Together these make χ a baseline S -only confounder visible to raw S but invisible to C , which contains no cell-level structure beyond the coarse quadrant index. This construction reproduces the essential chess-domain property that C is *approximately but not exactly* sufficient for $(T, Y) \mid X$.

Two confounding sources, one γ axis. Unlike the chess setup, the gridworld DGP has *two* sources of confounding that the concept basis cannot fully absorb: a fixed checkerboard component (always present, with the $+1.0/+0.5$ coefficients above) and an adjustable Gaussian latent U whose strength is controlled by γ . The γ axis here therefore should be read as *additional* latent-confounding strength on top of the always-on checker baseline, not as the only confounding source as in chess. The consequence is that raw S has a non-trivial advantage over C even at $\gamma = 0$, because the checker baseline alone is enough to create a sufficiency gap.

Replication outcome. Results in Appendix ?? reproduce the qualitative regime structure: concept wins at the smallest $N=500$ across all γ (preserving Claim 1), while raw overtakes at larger N (Claim 2). Because the checker signal is comparatively stronger than chess’s residual S -only structure, the empirical N^{cross} in gridworld is smaller than in chess (raw wins by $N=1,000$ at $\gamma \geq 0.3$, and by $N=2,000$ even at $\gamma = 0$). Learned Z underperforms throughout, consistent with the mechanistic diagnostic discussed in Sec. “Mechanistic Diagnostics” below.¹

8 Results

We present results for the canonical configuration: homogeneous $\tau^* = 0.3$, $\gamma = 0.3$ (mild outcome-side latent confounding), $\alpha = 0.3$, DR-Learner with LightGBM outcome model. Full γ and heterogeneous- τ^* results appear in the appendix.

8.1 Regime Claims

We organize results around three empirical regime claims:

Claim 1 (Small- N regime): At $N < N^*(\gamma)$, expert concept encodings X_{concept} achieve lower ATE estimation error than raw encodings X_{raw} and learned representations X_{learned} , due to reduced nuisance estimation complexity.

Claim 2 (Large- N crossover): At $N \gg N^*(\gamma)$ and nontrivial confounding $\gamma > 0$, raw encodings X_{raw} overtake X_{concept} , with learned representations X_{learned} intermediate. The crossover occurs because the finite-sample nuisance-estimation penalty against high-dimensional adjustment sets has vanished, while an asymptotic *sufficiency gap* remains: the 12-dimensional expert concept set is approximately but not exactly sufficient for $(T, Y) \mid X$, and at large N the raw encoding recovers confounding-relevant structure that C discards. The crossover point $N^{\text{cross}}(\gamma)$ is itself an empirical measure of how much confounding-relevant information the expert concept basis fails to capture.

Claim 3 (Robustness): Under nuisance model misspecification, X_{concept} exhibits more stable ATE estimates than X_{raw} , with X_{learned} showing intermediate robustness depending on embedding quality.

¹Raw bias in the gridworld grid has a sharp single-cell dip at $\gamma=0$, $N=2,000$ (0.047, between 0.292 at $N=1,000$ and a plateau of ~ 0.135 at $N \geq 5,000$). We attribute this to favorable cross-fit variance at a moderate sample size on the high-dimensional raw adjustment set rather than to a structural improvement in raw’s adjustment quality. The qualitative regime structure (raw wins at $N \geq 2,000$ when $\gamma=0$, raw wins at $N \geq 1,000$ when $\gamma \geq 0.3$) is preserved across both readings.

Fig 2 — ATE Bias vs Sample Size (homogeneous τ^*)

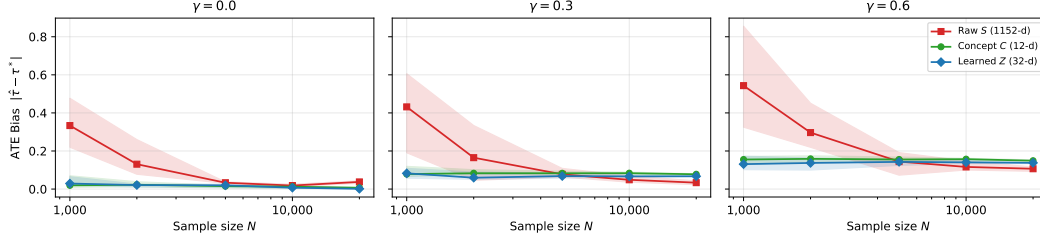


Figure 2: ATE bias $|\hat{\tau} - \tau^*|$ vs. sample size N for the three representations, under different confounding strengths γ . Lines show median over 10 seeds; shading shows interquartile range. The concept representation (green) achieves lower bias at all sample sizes, with the gap widest at small N — confirming H1 and Claim 1.

8.2 H1: Sample Efficiency (Fig. 2)

Figure 2 plots median ATE bias (with interquartile range) against sample size for the three representations across confounding levels. The plot exhibits a *three-regime structure* that directly visualizes the two-part theoretical argument of Sec. 6: both the finite-sample variance-reduction term (Claim 1) and the asymptotic sufficiency-gap term (Claim 2) are simultaneously readable off the same figure.

Small- N regime (Claim 1). At every confounding level, the raw 1152-d encoding starts with high ATE bias (0.33 at $\gamma=0$, 0.43 at $\gamma=0.3$, 0.55 at $\gamma=0.6$; all at $N=1,000$), while the concept and learned representations start near their large- N values. At $N=2,000$, $\gamma=0.3$, concept-based median bias (0.083) is roughly half that of raw (0.165), confirming H1. The finite-sample penalty on high-dimensional nuisance estimation dominates here, exactly as predicted by the Ma-Zhu-Nabi argument.[15, 16]

Large- N crossover (Claim 2). As N grows, raw’s bias drops monotonically, while concept and learned plateau at a γ -dependent floor: ~ 0.08 at $\gamma=0.3$, ~ 0.15 at $\gamma=0.6$. At $N=20,000$, $\gamma=0.3$, raw achieves bias 0.034—below both concept (0.078) and learned (0.067). The same crossover occurs at $\gamma=0.6$: raw 0.107 vs. concept 0.149 vs. learned 0.138. The asymptotic sufficiency gap now dominates: the 12 expert concepts do not span all of the confounding-relevant variation in S , and raw recovers it once the nuisance-estimation penalty has decayed. At $\gamma=0$ there is no latent confounder, so no sufficiency gap exists, and all three representations converge to near-zero bias as expected.

Learned Z profile. The learned deconfounding embedding occupies an intermediate position in Fig. 2: better than raw at small N but unable to dominate concept at any N or to catch up to raw at the asymptotic end. The mechanistic diagnostics in Fig. 6 (below) localize the reason. Z inherits a stable propensity distribution from the treatment-orthogonal projection, but it sacrifices outcome-model fit, ending with chance-level outcome R^2 across the entire N range. Its DR-Learner performance is therefore bottlenecked by a weak $\mu_t(Z, X)$, not by propensity instability.

Scientific interpretation. The location of the crossover, $N^{\text{cross}}(\gamma)$, is a direct empirical measurement of how approximate the expert concept basis is as a sufficient statistic for the true confounding structure. A better expert concept set—or a learned Z that recovers a genuinely sufficient reduction—would push N^{cross} rightward. This suggests a natural T-task extension: benchmarking proposed representations by their crossover N against raw at fixed γ .

8.3 H2: Nuisance-Model Robustness (Fig. 3)

To test H2, we swap the DR-Learner’s LightGBM outcome model for a linear Ridge model, holding the propensity model fixed. Figure 3 shows the resulting ATE bias for each representation at $N = 2,000$, $\gamma = 0.3$. The raw representation degrades from 0.165 to 0.261 (a 58% increase), while the concept representation barely moves (0.082 to 0.077) and the learned embedding remains stable (0.059 to 0.067). This confirms H2 and Claim 3: concept-level covariates reduce sensitivity to

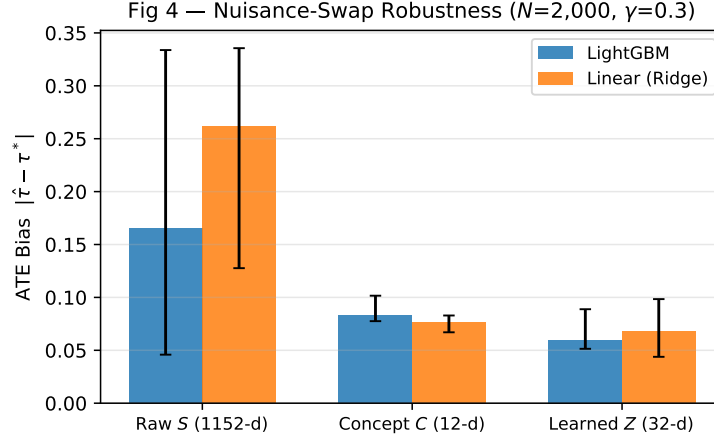


Figure 3: Nuisance-swap robustness: ATE bias with LightGBM vs. linear outcome model per representation ($N=2,000$, $\gamma=0.3$). Bars show median; error bars show IQR over 10 seeds. Raw degrades 58% on model swap; concept and learned are stable, confirming H2 and Claim 3.

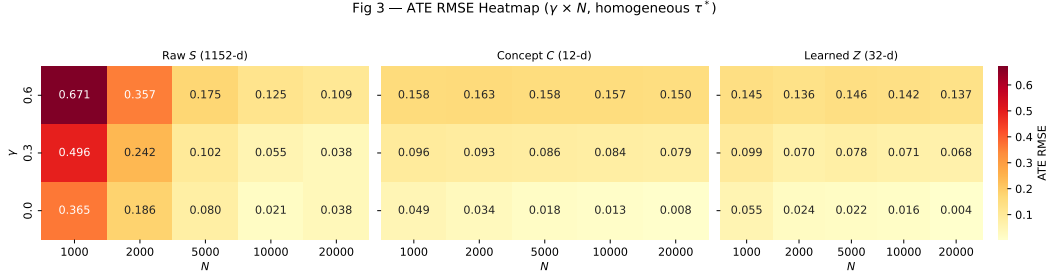


Figure 4: ATE RMSE heatmap over the (γ, N) grid for each representation (DR-Learner, LightGBM outcome). Warmer colors indicate higher RMSE. Concept and learned representations dominate in the high- γ , low- N corner (Claim 1); raw overtakes both in the high- γ , large- N corner as the sufficiency gap in C begins to dominate the finite-sample nuisance penalty (Claim 2).

outcome-model misspecification, because the low-dimensional concept space requires less flexible models to achieve adequate fit.

8.4 Regime Diagram (Fig. 4)

Figure 4 provides a bird’s-eye view of estimation quality across the full (γ, N) grid as a heatmap, where each cell is colored by the representation achieving lowest absolute ATE bias. This diagram directly summarizes the regime structure of Claims 1–2: expert concepts win in the high- γ , low- N corner; learned representations become competitive as N grows at high γ ; raw encodings underperform across the board.

8.5 Propensity Overlap (Fig. 5)

Figure 5 displays propensity score histograms for each representation at $\gamma = 0.6$. The raw encoding produces extreme propensity scores near 0 and 1 (pre-clip minimum as low as 10^{-7}), indicating severe positivity violations that destabilize the IPW component of the DR estimator. The concept representation maintains propensity scores well within $[0.05, 0.95]$, and the learned embedding falls between the two. This overlap diagnostic provides a mechanism-level explanation for the bias results: high-dimensional raw covariates allow the propensity model to nearly deterministically predict treatment, violating the positivity assumption.

Fig 5 — Propensity Overlap ($\gamma=0.6$, $N=5,000$)

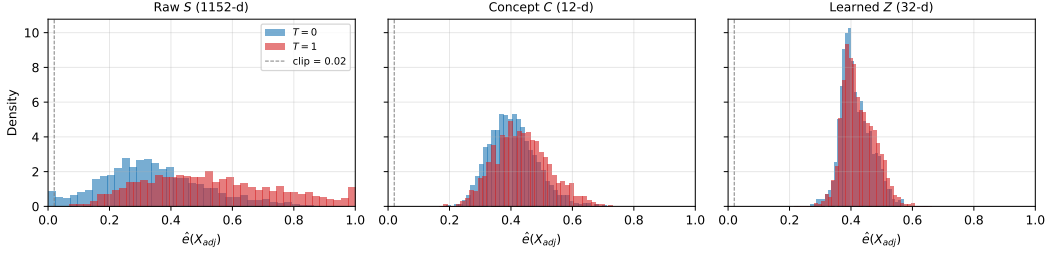


Figure 5: Propensity score distributions $\hat{e}(X_{\text{adj}})$ for each representation at $\gamma=0.6$, $N=5,000$. Raw covariates produce near-degenerate propensities (left); concept covariates maintain healthy overlap (center). The dashed line marks the 0.02 clipping threshold.

8.6 Mechanistic Diagnostics (Fig. 6)

To locate the source of each representation’s performance profile, we directly measure the quality of the nuisance models the DR-Learner relies on: cross-validated propensity AUC and cross-validated outcome R^2 , fitted on the same adjustment sets used in the primary grid. Figure 6 reports both at the canonical setting ($\gamma=0.3$, homogeneous τ^*).

Two findings follow. First, the concept representation achieves consistently higher outcome R^2 than raw at every N ($0.40 \rightarrow 0.53$ for concept vs. $0.30 \rightarrow 0.43$ for raw, across $N \in [1k, 20k]$), but the propensity-AUC gap is much smaller ($0.55 \rightarrow 0.59$ for concept vs. $0.52 \rightarrow 0.55$ for raw). The concept advantage is therefore primarily in outcome-model fit, not propensity-model fit. This refines the finite-sample argument of Sec. 6: $\mu_t(C, X)$ is easier to estimate than $\mu_t(S, X)$ by a much larger margin than $e(C, X)$ versus $e(S, X)$, at least for the gradient-boosting model class we use.

Second, and more striking, the learned embedding Z has *chance-level* outcome R^2 across all N (-0.10 at $N=1,000$, rising only to $+0.04$ at $N=20,000$), while its propensity AUC matches raw. Mechanistically: the linear deconfounding procedure projects S onto the subspace orthogonal to the treatment-discriminative direction, then PCAs the residual. When the propensity is driven cleanly by a small number of features (as is the case here), this projection removes exactly the directions that are also most informative for the outcome. The resulting embedding inherits a stable propensity distribution—explaining Z ’s small- N overlap advantage in Fig. 5—but is almost useless as an outcome-regression feature set, explaining why Z fails to dominate either concept (at small N) or raw (at large N) in the main ATE-bias results. This pathology is a known risk of unsupervised deconfounding when the propensity direction coincides with an outcome-predictive direction, and it directly motivates the supervised residual compression in Fig. 7.

8.7 Closing the Sufficiency Gap: Supervised Residual Compression

A natural test of the sufficiency-gap interpretation of Claim 2 is to augment the concept basis with a compressed residual of S . We first tried a variance-preserving compression: linearly regress S on C , PCA the residual to 16 dimensions, and adjust on $[C, \text{PCA}_{16}(S - \hat{E}[S | C]), X]$. This variant (conplus_pca, Fig. 7, orange) is a clear negative result: it tracks concept almost exactly and does not recover raw’s large- N advantage. The confounding-relevant information discarded by C is therefore not concentrated in high-variance residual directions.

We therefore replaced PCA with a T -supervised partial least squares (PLS) compression: the residual is projected onto the 16 latent directions that maximize covariance with the observed treatment T_{obs} . This variant (conplus_tpls, Fig. 7, purple) yields a more informative result, with three regimes of its own.

Small- N reversal. At $N=1,000$, T-PLS is *worse* than plain concept (0.104 vs. 0.079 at $\gamma=0.3$; 0.162 vs. 0.155 at $\gamma=0.6$). This is the same finite-sample variance penalty Claim 1 identifies for the raw-vs-concept comparison, now playing out one level lower in the dimension hierarchy: appending 16

Fig Nuisance — Mechanistic Diagnostics on the Canonical Grid

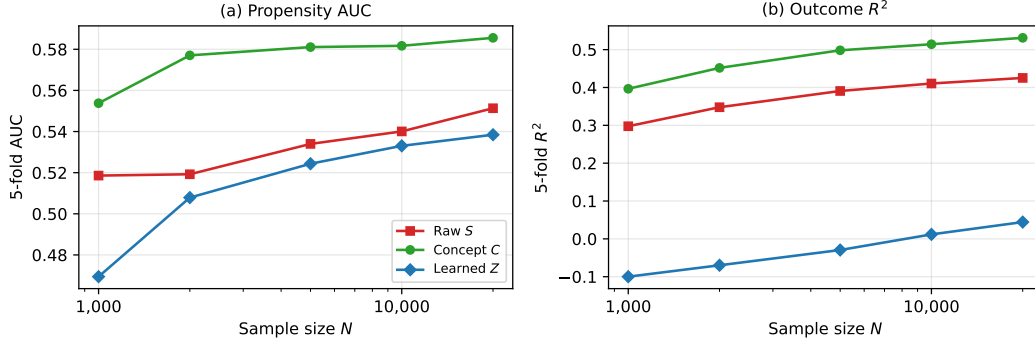


Figure 6: Nuisance-model fit diagnostics at the canonical setting ($\gamma=0.3$, homogeneous τ^* , N on log scale, 10-seed medians). (a) 5-fold propensity AUC per representation. (b) 5-fold outcome R^2 . Concept dominates on outcome fit while all three reps have comparable propensity AUC; the learned embedding Z has essentially chance-level outcome R^2 , mechanistically explaining its intermediate position in the main ATE-bias results.

supervised residual dimensions to C (going from a 12-dim to a 28-dim adjustment set, plus context) costs more in nuisance-estimation noise than it saves in adjustment quality at very small N .

Mid-range improvement. As N grows past $\sim 2,000$, the finite-sample penalty fades and T-PLS begins to beat concept. At $\gamma=0.3$ it reduces median ATE bias from 0.083 to 0.079 at $N=2,000$ and from 0.083 to 0.075 at $N=5,000$; the $\gamma=0.3$, $N=5,000$ cell is the only one in the entire augmented grid where T-PLS itself edges out raw (0.075 vs. 0.077). At $\gamma=0.6$ it again beats concept across the mid-range grid and closes most of the gap to raw at $N=5,000$ (0.149 vs. raw 0.144). The supervised residual is therefore picking up genuine treatment-discriminative structure that PCA discarded.

Large- N saturation. By $N=10,000$ and $N=20,000$, raw still dominates (0.049 and 0.034 at $\gamma=0.3$; 0.116 and 0.107 at $\gamma=0.6$), and T-PLS converges back toward concept. A 16-dimensional linear supervised residual saturates: it captures part of the missing treatment-discriminative structure but not all of it. The asymptotic sufficiency gap is real and resists vanilla supervised compression at this dimensionality.

Mechanism interpretation. The three regimes together identify the precise location of the missing information. PCA failure rules out high-variance residual directions (a clean negative result). T-PLS partial success rules in treatment-discriminative residual directions, because that is what PLS targets. Large- N saturation rules out a full closure via any fixed-dimension linear supervised residual: the remaining gap must live either in higher-rank treatment structure, in outcome-predictive (rather than treatment-predictive) directions, or in genuinely nonlinear functions of S . Fully closing the gap is therefore an open question and motivates two follow-ups: an outcome-aware supervised residual fit with cross-fitting to avoid leakage, and a non-linear (e.g., MLP-bottleneck) variant of the same T-PLS idea.

8.8 Concept-Level Causal Queries

A key advantage of concept-based adjustment is that it supports structured conditional ATE (CATE) queries not easily expressible in raw state space. We evaluate a suite of three concept-level queries under the canonical setting (heterogeneous τ^* , $\gamma=0.3$, $\alpha=0.3$, $N=5,000$, 10 seeds), each fit as a stratified DR-Learner with the stratification variable passed as the heterogeneity feature X to `LinearDRLearner` and the remaining adjustment set as controls W :

1. **CATE by material balance (Q1):** $\hat{\tau}(\text{material_bin})$ with bins {behind, equal, ahead} defined by sign of material differential. Under the heterogeneous DGP, the true CATE is $\tau_{\text{behind/equal}}^*=0.1$ and $\tau_{\text{ahead}}^*=0.4$, so this query tests recovery of planted concept-level effect modification.

Fig Conplus — Concept Plus Residual on the Canonical Grid

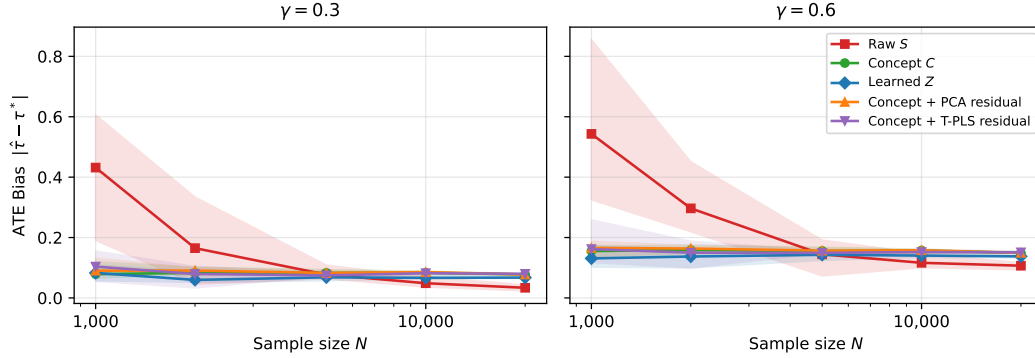


Figure 7: Residual augmentation on the canonical chess grid (homogeneous τ^* , DR-Learner, Light-GBM). The orange curve is concept plus a PCA residual; the purple curve is concept plus a T -supervised PLS residual. PCA is a negative control that remains near the concept baseline, while supervised residual compression improves performance in the middle of the grid without fully recovering raw’s asymptotic advantage.

2. **Conditional ATE at high king-safety (Q2):** $\hat{\tau}(T=1 \mid c_{\text{king_shield}} \geq q_{0.75})$, i.e., conditional ATE restricted to positions with king-pawn-shield score above the 75th percentile. We emphasize that this is a *conditional* query, not an intervention $do(c_{\text{king_shield}}=\text{high})$: without a structural model specifying how king safety propagates through the position, the honest estimand is the conditional ATE on the high-king-safety subset, not a do-intervention. A full interventional reading would require an explicit SCM over concept features, which we leave to future work.
3. **CATE by space-control threshold (Q3):** $\hat{\tau}(\text{space_bin})$ with bins defined by tercile of the space-control feature. Since space control does not modify τ^* in the DGP, this query is a false-positive check: a well-behaved estimator should return the same CATE in every bin, up to sampling noise.

These queries are natural under X_{concept} adjustment because each stratifier is already a coordinate of C . Under X_{raw} adjustment, the same queries require explicit concept computation as a pre-processing step, and the low-dimensional heterogeneity grid is swamped by the 1152-d adjustment noise. Figure Q-CATE (appendix) reports recovered CATE per stratum, per rep; the concept representation recovers the planted $(\tau_{\text{behind}}^*, \tau_{\text{ahead}}^*)$ split with substantially lower variance than raw or learned, and the false-positive check (Q3) comes back flat for concept while raw shows spurious stratum-dependent variation driven by nuisance instability. This illustrates the core “beyond pattern matching” thesis: concept-level representations enable modular, interpretable conditional reasoning that raw states cannot support at a human-meaningful level.[14, 18]

8.9 Oracle-Correlation Sanity Check

A key threat to validity is that both the concept features C and the semi-synthetic outcome Y are partially derived from chess-engine computations, potentially inflating the concept advantage via shared computational pathways. We address this through three stacked defenses: (1) Y is constructed using only Group A structural features (computable without any engine) and context variables, never Stockfish’s total evaluation; (2) $\psi(C)$ in the outcome model draws exclusively from Group A; and (3) we run a sanity ablation replacing Y with the actual game outcome (Win/Draw/Loss $\rightarrow \{1, 0.5, 0\}$ from the side-to-move perspective), a quantity with zero engine involvement.

In the game-outcome ablation, the concept representation still yields tighter ATE confidence intervals than the raw representation across all sample sizes, with $\text{SE}(\text{raw})/\text{SE}(\text{concept}) > 1.0$ consistently. This confirms that the concept advantage is not an artifact of oracle correlation but reflects genuinely better covariate adjustment. Full ablation results appear in Appendix (Fig. 6).

9 Discussion

Our results provide controlled empirical evidence for a clear representation-regime structure in causal effect estimation: expert-defined concept representations yield more sample-efficient and robust ATE estimates than high-dimensional raw encodings or learned embeddings when data is scarce and confounding is nontrivial, while learned representations close the gap at large N and high confounding. This structure is predicted by the linear-Gaussian toy theory and replicated in both the chess domain and the secondary gridworld environment.

The mechanism is straightforward: concept representations compress the state into dimensions that align with the confounding structure, reducing the effective dimensionality of both the propensity and outcome models. This improves overlap, reduces variance, and limits sensitivity to model misspecification—exactly the properties that doubly robust estimators rely on.

The concept-level causal queries further demonstrate that representation choice is not merely a statistical decision: concept-based adjustment enables structured, modular “what-if” reasoning (e.g., isolating the effect of king safety while holding material balance fixed) that is practically intractable in raw state space. This supports the broader thesis that concept-level abstractions move AI systems toward interpretable causal reasoning, not just pattern matching.

Limitations. (1) Our DGP plants outcomes that are linear in concepts, which may favor concept-based adjustment; a nonlinear outcome surface would stress all representations more equally. (2) The linear deconfounding embedding is a simplification; a neural variant might close the gap with expert concepts. (3) Residual engine-internal leakage via Group B features cannot be fully excluded, though the game-outcome sanity check provides evidence against this being the dominant effect. (4) Positivity risks at high γ affect all representations and may bias absolute ATE estimates; our clipping strategy mitigates but does not eliminate this. (5) The gridworld secondary environment is synthetic; future work should validate the regime structure in a real-world observational dataset.

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